

Scalable Continuous Synthesis of Organozinc Reagents and Their Immediate Subsequent Coupling Reactions

Gabriele Menges-Flanagan,* Eva Deitmann, Lars Gössl, Christian Hofmann, and Patrick Löb



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ABSTRACT: The continuous synthesis of organozinc reagents and their immediately following subsequent also continuous consumption in catalyzed and noncatalyzed coupling reactions were investigated. In the first step, a bed of Zn turnings at variable liquid throughputs and concentrations of organic halide solutions was used, and the formed Zn organometallics were analyzed for quality control. They were then directly pumped into a second step, namely, Reformatsky, Saytzeff, and Negishi coupling reactions. In the organozinc halides' formation, a novel process window was employed by using a large molar excess of Zn turnings and investigating mechanical as well as chemical Zn activation. Subsequent couplings of the freshly prepared Zn organometallics were done using examples of a Reformatsky, Saytzeff, and Negishi coupling reaction. For the Zn organometallics' formation, a laboratory-scale reactor setup previously built for Grignard reagent formation was evaluated including a Zn replenishing unit; the same reactor was also used in the metal-catalyzed subsequent step (Negishi coupling). The main objective of this work was to establish the scalable continuous formation of Zn organometallic reagents enabling fast and safe process optimization, analyze the reagents for their purity, and then immediately consume them in various follow-up steps, always only leaving a very small amount of reactive and sensitive organometallic reagent in the setup. It was found that full conversion of the employed halides could be achieved within a single passage through the reactor with organozinc yields of 82–92%, as well as being able to successfully perform subsequent non- and metal-catalyzed coupling steps with yields of up to 92%. A pilot-scale setup allowing a liquid throughput of up to 3–5 L/h has also been built and is ready to be tested with the synthesis as established here.

KEYWORDS: continuous processing, flow chemistry, organometallics, scale-up, organozinc compound

INTRODUCTION

Organozinc reagents were among the initial organometallic species to be discovered and used in C–C bond formation reactions. The first formation of a zinc reagent was performed by E. Frankland in 1848, reacting zinc metal with ethyl iodide.¹ They enjoyed prominence and were used for C–C bond formation in the second half of the 19th century until the discovery of Grignard reagents by Victor Grignard,² and their subsequent coupling reactions in the actual Grignard reaction eclipsed their use. However, through their combination of a lower reactivity compared to Grignard reagents and therefore a higher tolerance toward functional groups (esters, ketones, cyanides, halides, primary and secondary amines, etc.), within reaction partners they constitute ideal reagents for multistep synthesis particularly in the synthesis of active pharmaceutical ingredients (APIs).³ The organozinc reagents as described in this paper of the general formula RZnX (R constituting an organic moiety and X denoting a halide, most often Cl or Br) can be formed either via direct insertion of metallic zinc into an organo-halide (RX) bond or via transmetalation.^{4–6} Newer developments in the field of synthesizing Zn organometallic reagents have been concerned with the improvement of their handling and particularly their air and moisture sensitivity, where the reagents can be stabilized as solid-form Zn pivalates which can then more easily be handled and used in follow-up coupling reactions.^{7,8} As mentioned above, organozinc halides

can be employed as reactive intermediates in complex multistep synthetic processes. Among others, they are utilized in non-metal-catalyzed Reformatsky reactions (condensation reaction between a carbonyl compound and an α -halide carboxylic acid ester),^{9,10} Saytzeff reactions (reaction between an active organozinc halide and either an aldehyde or ketone to form alcohols),¹¹ and Blaise reactions (formation of β -ketoesters via the reaction of Zn with an α -bromoester and a nitrile).^{12,13} The most prominent metal-catalyzed coupling reaction is the Negishi coupling between a zinc organometallic species and an organic halide under Pd or Ni catalysis, which earned its inventor the 2010 Nobel prize for chemistry, almost 100 years after Victor Grignard was awarded his for the equivalent Mg organometallic species.^{3,14} Their unique combination of above-mentioned functional group tolerance, moderate reactivity, and formation of nontoxic side products as compared to other organometallic reagents truly renders them ideal candidates for a multitude of C–C bond formation

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reactions and constitutes some crucial advantages over the use of other reactive intermediates.

In recent years, a number of studies have been published on a small laboratory scale synthesizing organozinc species continuously followed by their consumption in a second reaction step.^{7,15–21} These studies prove how valuable organozinc species are, particularly as intermediates for manufacturing in the pharmaceutical industry; however, their wider-spread utilization is hampered by their instability and sensitivity to oxygen and water. The results presented here demonstrate how a scalable continuous processing approach can lead to “on demand” production of organozinc species where these *in situ* generated reactive intermediates are immediately consumed in a second subsequent, continuous, and scalable processing step. The general interest in the topic of combining flow chemistry with the generation and use of organometallic reactive intermediates can also be seen in several recent reviews on the subject.^{22–24}

It is noteworthy here that, among the currently most used synthetic pathways in API synthesis, no new routes have been established within the last 20 years, and also within the 1980s and 1990s, only two new synthetic pathways (Suzuki–Miyaura and Buchwald–Hartwig) were established.²⁵ In an analysis by Roughley et al.,²⁶ it was found that roughly 10 different reaction types make up about two-thirds of the reactions used in API syntheses today. Since there may be more and more APIs synthesized but not necessarily more new drugs being approved, and a bias is apparent toward certain classes of reagents, there is a clear need to make new reaction pathways accessible that possess a high functional group tolerance and therewith eliminate the need for protecting groups, which is applicable to the Zn organometallic species investigated here.²⁵ The chemistry explored ideally combines well-established chemistry (e.g., Pd-catalyzed C–C cross couplings) with the wider accessibility of Zn organometallic reagents through continuous processing. Particularly, the Suzuki–Miyaura coupling is an impressive example of how the uptake of novel chemistry (here for the synthesis of biphenyls) can lead to a steady increase in using such synthetic fragments from 2% in the 1980s to 12% today.²⁵ This paper describes an expansion of the playing field for process chemists in the pharmaceutical industry through providing scalability of the continuous processing of organozinc species to industrially relevant throughputs, with technology ready to be adopted in industry. More recently, a review on “the importance of synthetic chemistry in the pharmaceutical industry” found that, despite the advent of new pathways such as, e.g., biocatalysis, olefin metathesis, and cross-couplings, there remains a need for innovation in synthetic chemistry.²⁷

RESULTS AND DISCUSSION

For the synthesis of RZnX reagents, a laboratory-scale reactor as designed and manufactured via 3D laser melting by Fraunhofer IMM for Grignard reagent formation²⁸ was used. For the case of filling with Mg turnings, a behavior close to plug-flow was found (Bodenstein numbers of up to 93 at flow rates between 1 and 4 mL/min), and a similar behavior is expected here; however, it has not been investigated in detail for Zn granules. The same reactor was used in the subsequent palladium-catalyzed Negishi coupling, while the noncatalyzed Reformatsky and Saytzeff reactions were performed with a PTFE coil immersed in a thermostat bath to provide sufficient residence time for the synthesis in question. Manufacturing the

first step reactor via 3D laser melting allows cost-efficient reactor fabrication while enabling access to heat exchanger structures not amenable via conventional manufacturing, which plays a crucial role in accomplishing effective heat exchange structures for scale-up.²⁹ Figure 1 details the reactor

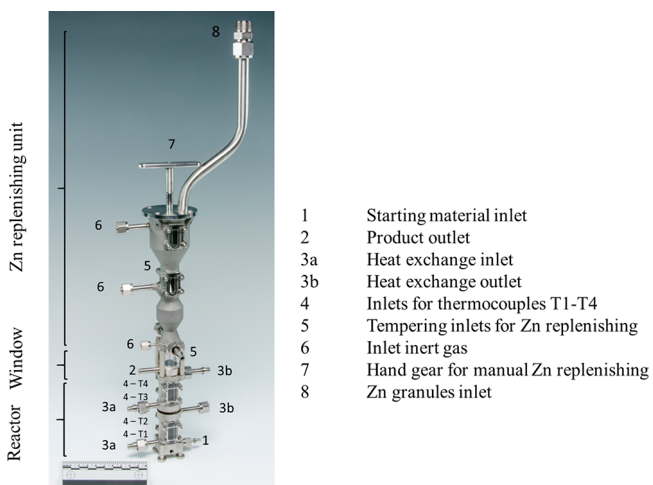


Figure 1. Laboratory-scale reactor for the synthesis of organometallic reactive intermediates via solids processing. Used with permission from ref 28. Copyright 2020 American Chemical Society.

constructed and patented³⁰ initially for Grignard reagent formation and used here for organozinc reagent formation. It allows for two different temperature zones to be applied within a synthesis since the reactor is manufactured from two equal halves laser welded together. Furthermore, it contains a manual Zn replenishing unit that can be attached to the main reactor body.

Titration was used as an immediate tool to determine RZnX concentration following the procedure described by Huck¹⁵ with a modification for the analysis of allyl zinc bromide using a back-titration as performed by Knochel³¹ and specifically adapted by Deitmann.³² For a more detailed analysis of the remaining starting material and possible side product content, the samples were quenched with 1-butanol and analyzed via gas chromatography (GC). Initially, activation of the Zn was attempted solely through mechanical action through a jogging motor. However, it was found that full halide conversion could not be achieved using only mechanical activation. The start of Zn organometallic reagent formation could be observed but only occurred in the upper half of the reactor. Therefore, the reactor volume was not fully used for the formation of the organozinc species. This might be overcome by investigating the use of a more sophisticated mechanical activation directly within the reactor or the employment of larger Zn granules. In the following, chemical activation has been used for the employed Zn granules in the reactor, namely, following the procedure by Huck et al.¹⁹ using a mixture of chlorotrimethylsilane and 1-bromo-2-chloroethane.

All substances investigated in the organozinc reagent preparation were used at a concentration of 0.5 mol/L in THF since reference substances at this concentration are commercially available. Residence times in the reactor ranged from about 3.0 to 13.0 min and were dependent on the halides' reactivity and therewith time needed to reach full conversion within a single passage through the reactor. To establish optimal reaction conditions and ensure reproducibility,

multiple runs for each new RZnX reagent were performed. Investigated variables were reaction temperature (through the setting of the thermostat bath (TT) employed for heating/cooling the reactor) and halide flow rate with the goal, again, to achieve full halide conversion within a single passage of halide solution through the reactor. In other words, the goal was to achieve a maximum possible flow rate while simultaneously using minimal energy for excessive heating/cooling of the reactor. The organozinc formation reactions could be investigated with the reactor as given above over a period of up to 4 h, and the Zn replenishing unit could be tested for each organozinc reagent in a two-step process verifying its general applicability when running experiments over a longer period of time. Figure 2 shows the temperature profiles along the reactor for the synthesis of a 0.5 M benzyl zinc bromide solution in THF.

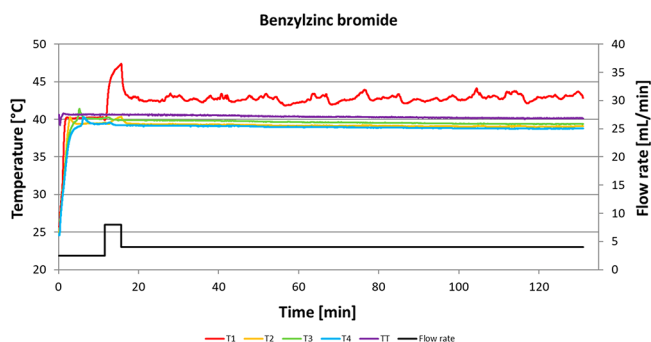


Figure 2. Temperature profile along the reactor. Flow rate, 4 mL/min (filling: 8 mL/min); 0.5 M benzyl Br in THF; thermostat bath setting (TT), 40 °C; full conversion of benzyl Br reached; yields, 90–92%.

It was found that the temperature profiles along the reactor are more regular compared to previously investigated Grignard reagents formed from Mg turnings in the same reactor setup and do not show hotspot formation, which can mainly be attributed to the Zn organometallics' lower reactivity. Table 1 summarizes the zinc organometallic reagent synthesis investigated detailing halide concentrations employed, maximum flow rate possible for full conversion, temperature setting of the thermostat bath, and yields of organozinc reagents achieved (determined by manual titration).

It was found that all reagents could be formed in good to very good yields, e.g., comparable to achievements reached by Huck et al.¹⁵ with very limited coupling of starting material and product (as investigated via GC analysis of quenched samples). This is particularly noteworthy for the benzyl and allyl zinc bromides since these tend to show a tendency to large amounts of coupling. In the experiments, the reactor was freshly filled with about 105 g of Zn granules each time which, using 0.5 molar starting solutions, corresponded to a molar excess of the metal compound of about 250 and an accessible free volume of about 13 mL within the reactor. Particularly for the case of

benzyl zinc bromide, the suppression of coupling between starting material and product is noteworthy; the coupling product lay below the detection limit of the GC method used and is a notable improvement to the batch synthesis where up to 100% coupling product under nonoptimized conditions is observed.³³ The reactions were performed over a time period of up to 4 h. In some cases, a small amount of white precipitate could be observed after the reaction within the reactor, most likely due to side product formation through reaction of organozinc species with traces of water. However, this did not impede experiments being run here and did not result in a clogging of the reactor. In the future, longer runs and great care to ensure the absence of water are needed to investigate the true potential of using this form of continuous processing over the course of a longer period of time. This will also clarify further the influence of the Zn bed changing over time, with particles shrinking in size and being partly eluded from the reactor.

Each of the above synthesized RZnX compounds was used in a specific coupling reaction (Reformatsky, Saytzeff, and Negishi coupling) using either a Teflon tubing coil (inner volume about 48 mL) as residence time loop immersed in a thermostat bath for the noncatalyzed reactions or a catalyst-filled stainless steel reactor identical in construction to the one used in the organozinc reagent formation. Figure 3 shows the

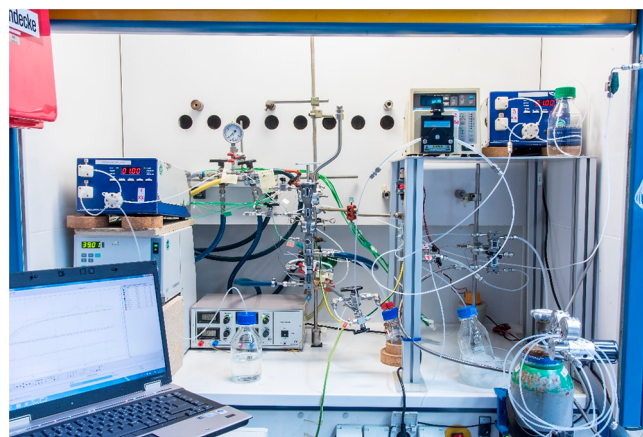


Figure 3. Laboratory setup for the synthesis and Negishi coupling of benzyl zinc bromide and 1-bromo-4-ethylbenzene.

setup as used for the 2-step synthesis of benzyl zinc bromide formation followed by the Negishi coupling of the formed organozinc halide with 1-bromo-4-ethylbenzene. It must be noted here that a flow rate of 4 mL/min could be used in the synthesis of benzyl zinc bromide alone. However, in the coupled reaction steps, the flow rate had to be reduced (1 mL/min), so a sufficient residence time for the second step over the catalyst bed could be achieved.

Table 1. Reaction Conditions and Yields Obtained for RZnX Reagents

Zn reagent	concentration starting material [mol/L]	flow rate [mL/min]	residence time [min]	temperature TT [°C]	concentration (achieved in flow) [mol/L]	yield [%]
Et Zn Br acetate	0.5	filling, 8; run, 2	6.5	40	0.42–0.44	84–88
benzyl Zn Br	0.5	filling, 8; run, 4	3.25	40	0.45–0.46	90–92
allyl Zn Br	0.5	filling, 4; run, 1	13	10	0.41–0.42	82–84

Figure 4 shows the temperature profiles obtained in the coupling of benzyl zinc bromide with 1-bromo-4-ethylbenzene,

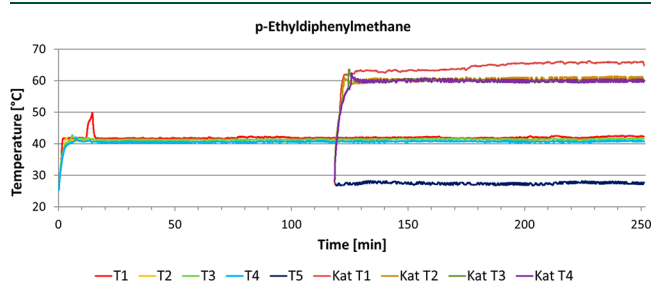


Figure 4. Temperature profile along the reactor for first step of benzyl zinc bromide formation and Negishi coupling with 1-bromo-4-ethylbenzene (T1–T4, temperatures along the reactor in first step; T5, temperature observed directly after mixing via a T-piece; Kat T1–Kat T4, temperatures along the 2nd step metal catalyst filled reactor).

showing the initial establishment of steady-state conditions for the first step (organozinc reagent formation) followed by the second metal-catalyzed Negishi coupling step.

Table 2 summarizes the results obtained in the three two-step syntheses performed. It is noteworthy here that in all cases an excess of organozinc reagent was used as following, e.g., the example by Huck et al.¹⁵ Furthermore, it was found that, due to the organozinc species' nature, a peristaltic pump rather than a syringe pump had to be used for dosing of the reactive intermediate.

It was found that the best results could be achieved for the Negishi coupling employing benzyl zinc bromide as well as for the Saytzeff reaction using allyl zinc bromide with yields around 90%, comparable to or even exceeding yields reported in the literature.^{16,34} The Reformatsky synthesis reacting ethyl zinc bromoacetate with 3-methoxyacetophenone posed more problems, and the excellent yields achieved by Berton et al. on a smaller-scale continuous process¹⁹ could not be reproduced in the setup described here. This is associated with an incomplete conversion of the carbonyl compound (3-methoxyacetophenone); a maximum of 43% was converted during the experiments even when the residence time was tripled by using an additional stainless-steel coil as residence time loop. The low yield observed for this particular synthesis needs to be further investigated and is potentially due to gas formation through local hotspots within the reactor. However, the general applicability of the two-step synthesis forming a reactive intermediate *in situ* and converting it in a follow-up step could be shown.

In order to achieve a scale-up of the two-step synthesis as described above, a pilot-scale reactor setup has been designed and constructed, and initial tests have been performed for the Grignard reagent formation. Figure 5 shows a pilot-scale



Figure 5. Pilot-scale reactor established for the synthesis of organometallic species to be tested for the scale-up of organozinc species investigated here.

reactor used when piloting the formation of a Grignard reagent at a throughput of 2.7 L/h. This setup is ready to also be tested for the synthesis of organozinc reactive intermediates and can be extended to include a second step with pilot-scale throughputs.

Table 2. Reaction Conditions and Yields Obtained for RZnX Reagent Formation and Subsequent Couplings

Zn reagent	carbonyl compound 2nd step	carb. compound [mol/L]	reaction type	product	total flow rate [mL/min], 1:1 ratio	residence time 2nd step [min]	temp. [°C] 1st/2nd step	conversion of starting materials [%] 1st/2nd step	yield, 2nd step [%]
Et Zn Br	3-methoxyacetophenone	0.15	Reformatsky	ethyl-3-hydroxy-3-(3-methoxyphenyl) butanoate	4	12	40/60	100/33–47	38
benzyl Zn Br	1-bromo-4-ethylbenzene	0.3	Negishi	p-ethyldiphenylmethane ^a	2	6	40/60	100/100	92
allyl Zn Br	cyclohexanone	0.35	Saytzeff	1-allyl cyclohexanol	2	24	10/18	100/100	88

^aNegishi coupling performed in a reactor similar in construction to the one for the first step; catalyst used was SiliaCat DPP-Pd with a 0.3 mmol/g catalyst loading.

■ EXPERIMENTAL SECTION

Typical Flow Procedure for Zn Organometallics Synthesis. The reactor was charged with about 105 g of zinc turnings while the jogging motor at the bottom of the reactor was already started (6 V). Argon was used for inertization, and four thermocouples were placed along the reactor's Zn bed for temperature monitoring. The activation of the zinc turnings was carried out by pumping a solution of 0.6 mol/L chlorotrimethylsilane and 0.24 mol/L 1-bromo-2-chloroethane in dry THF with a flow rate of 2 mL/min and a thermostat bath temperature of 40 °C through the zinc bed for about 10 min. After activation was done, the reactor was tempered using a thermostat bath temperature suitable for the desired RZnX synthesis and flushed with the halide starting material in dry THF at an elevated flow rate for up to 10 min, resulting in a temperature increase as a hint of the initiation of the exothermic reaction. The flow rate was then reduced to keep the temperature below the boiling point of the solvent, and the product solution was collected. Samples of the product solution were taken for gas chromatography analysis and titration to determine the organozinc halide concentration. Specific reaction conditions for the investigated organozinc halide reagents are shown in Table 1.

Typical Flow Procedure for Zn Organometallics Coupling Reactions. The organozinc halide, which was produced in the previous reaction step, was pumped from its collection vessel through an in-line filter with ceramic fibers and to the subsequent synthesis step by means of a peristaltic pump. In a T-piece, the organozinc halide was contacted with the electrophilic substrate (in dry THF), and the reaction mixture was further fed into the reactor (PTFE coil or stainless-steel reactor as in the first step). Temperature control was maintained using a thermostat bath, and the outlet solution leaving the reactor was collected in an ammonium chloride solution cooled with ice. Samples of the outlet solution were analyzed by means of gas chromatography. Specific reaction conditions are shown in Table 2. In the case of *p*-ethyl diphenylmethane synthesis, the reactor contained 5 g of the catalyst SiliaCat DPP-Pd mixed with glass beads (used for volume reduction).

Processing of Collected Samples. After quenching with ammonium chloride solution in the previous synthesis step, precipitated salts were filtered off, and the organic phase was extracted in the separating funnel by means of a suitable solvent (1-allylcyclohexanol: diethyl ether, *p*-ethyl diphenylmethane and ethyl 3-hydroxy-3-(3-methoxyphenyl)-butanoate: ethyl acetate). The organic phase was dried using sodium sulfate; the sodium sulfate was filtered off, and the organic phase was concentrated on the rotary evaporator. The concentrated product was again transferred to a separating funnel with solvent. A white precipitate was formed that could be transferred to the aqueous phase by adding some distilled water. Acidification of the aqueous phase by a few drops of 10% hydrochloric acid dissolved the precipitate so that the aqueous and organic phases could be separated in the separating funnel. The ether phase was again concentrated on the rotary evaporator to obtain the crude product (yield given as determined by GC analysis).^{16,19,34}

■ MATERIALS AND METHODS

All solvents and reagents were purchased from commercial suppliers and were used without further purification: allyl

bromide (Acros Organics, 99%), 0.5 M allyl zinc bromide in THF (Alpha Aesar), benzyl bromide (Thermo Fischer Kandel, 99%, stabilized), 0.5 M benzyl zinc bromide in THF (Sigma-Aldrich), 1-bromo-2-chloroethane (Merck, ≥98%), 1-butanol (Sigma-Aldrich, 99.7%), chlorotrimethylsilane (Sigma-Aldrich, ≥98%), cyclohexanone (Alfa Aesar, ≥99%), ethylbromoacetate (Merck, >98%), iodine (Fluka, >99.5%), 3-methoxyacetophenone (Acros Organics, 98%), SiliaCat DPP-Pd heterogeneous catalyst (Silicycle Inc.), THF (T H Geyer, 99.9%, water-free), and Zn turnings (Sigma-Aldrich, −30mesh).

Postnova syringe pumps (PN 1610 syringe dosing system) were used for the first step of the RZnX formation and to pump the carbonyl compound in the second step. For dosing the freshly prepared organozinc reagent, a peristaltic pump Masterflex L/S using PTFE tubing from Cole Palmer was used. The reactors were tempered using thermostatic baths from Julabo (F31-C) and Peter Huber Kältemaschinenbau (CC-405, CC-3). The temperatures were measured using miniature thermocouples (type K from RS Components), recorded by a data logger (Type Expert Key 200L, from Delphin Technology), and for evaluation transferred to a notebook. Titration of the Zn reagent was done following the method of Berton et al.¹⁹ and an adapted back-titration as described by the Knochel group.³⁰ Analysis for the RZnX compounds was done via quenching the samples with 1-butanol and analyzing via calibrated GC measurements, after workup crude products from the investigated second steps were also analyzed via GC. A Varian GC 3900 system with Varian 8400 GC-autosampler was used for GC analysis.

■ CONCLUSIONS

In conclusion, a continuously operating two-step laboratory setup for organozinc reagent formation and immediate consumption in subsequent catalyzed and noncatalyzed coupling reactions was established and tested. It was found that mechanical activation within the reactor alone did not create sufficient Zn surface activation and only yielded incomplete halide conversion, so chemical Zn activation had to be employed. However, for each organozinc species, full halide conversion in one single reactor passage could be achieved in good to very good yields with very little coupling product formation observed via quenching and GC analysis of samples. The reactor was operated at a large Zn excess (about 250-fold), and consumed Zn could be refilled via a metal replenishing unit attached to the reactor. Reformatsky, Saytzeff, and Negishi coupling reactions were performed using *in situ* generated Zn organometallic species. For the examples of the Saytzeff reaction and Negishi coupling, good yields could be obtained. In the case of the Reformatsky synthesis, full conversion of the employed carbonyl compound could not be achieved, and gas formation was observed during the reaction. This synthesis needs to be re-evaluated since a similar process by Berton et al. on a smaller throughput scale resulted in excellent yields.¹⁹ A scaled-up modular setup to pilot-scale throughputs (scale-up factor of 15) employing the same principle of using a large molar excess of the Zn within the reactor and also including a truly continuous Zn replenishing unit was designed and manufactured and is ready to be tested for the formation of organozinc species.

AUTHOR INFORMATION

Corresponding Author

Gabriele Menges-Flanagan – Fraunhofer IMM, 55129 Mainz, Germany; orcid.org/0000-0002-8769-7247; Phone: +49 6131 990425; Email: gabriele.menges-flanagan@imm.fraunhofer.de; Fax: +49 6131 990205

Authors

Eva Deitmann – Fraunhofer IMM, 55129 Mainz, Germany; Fachhochschule Münster, 48565 Steinfurt, Germany
Lars Gössl – Fraunhofer IMM, 55129 Mainz, Germany; Hochschule Darmstadt, 64295 Darmstadt, Germany
Christian Hofmann – Fraunhofer IMM, 55129 Mainz, Germany
Patrick Löb – Fraunhofer IMM, 55129 Mainz, Germany

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.oprd.0c00399>

Notes

The authors declare no competing financial interest.

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